BETWEEN THE LINES: RHETORICAL ANALYSIS OF OFFICIAL PUBLIC DISCOURSE ON THE 2003 INVASION OF IRAQ

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Topic Introduction

The attacks of September 11, 2001 brought urgent attention to minimizing the threats posed by terrorist groups in the Middle East. Many feared that Iraq could supply terrorists with weapons of mass destruction (WMD), a term referring to nuclear, biological, and chemical weapons that have devastating effects. While the United Nations had urged Iraq to dismantle said weapons programs during the Gulf War, the George W. Bush administration contended that Saddam Hussein, the Iraqi president, had continued development. The U.N. did not authorize an invasion, but the U.S. formed a "coalition of the willing" to topple Saddam Hussein and ensure the cessation of any programs for weapons of mass destruction. Following the invasion, investigations reported no evidence of active WMD programs. While the wider Iraq war covered vast and complicated territory, the initial claims of WMD are widely recognized as false and considered a failure within the Intelligence Community (Council on Foreign Relations).

Several official reports released in the years immediately following the invasion attribute the inaccuracies in WMD analysis to a widespread overstatement of confidence. A direct quote from The Commission on the Intelligence Capabilities of the United States Regarding Weapons of Mass Destruction (2005) details this concern:

> "We conclude that the Intelligence Community was dead wrong in almost all of its pre-war judgments about Iraq's weapons of mass destruction. This was a major intelligence failure. Its principal causes were the Intelligence Community's inability to collect good information about Iraq's WMD programs, serious errors in analyzing what information it could gather, and a failure to make clear just how much of its analysis was based on assumptions, rather than good evidence. On a matter of this importance, we

simply cannot afford failures of this magnitude" (Commission on the Intelligence Capabilities).

The leading figures in public discourse have distinct speaking roles, with differing linguistic registers that are critical for the analysis of decision-making (Jervis). In this case, the central question is whether natural linguistic profiles show significant trends across speakers and roles. Even an individual word can have a significant impact on an audience's understanding of the threat landscape. For example, the Butler Report on British performance's investigation into WMD claims found that linguistic standards for conveying certainty were insufficient, with terms such as probable and likely used interchangeably despite differing intended meanings (Jervis 127).

Research Questions

This project hypothesizes that public-facing executive communication in the lead-up to the Iraq War produced measurable convergence in linguistic styles across significant time periods and institutional roles. Drawing on LIWC analysis of select documents, it further investigates whether these linguistic profiles can predict the later-determined accuracy of the claims made.

It is important to specify that the aim of this project is not to determine the cause of inaccurate statements or to indicate deception, but rather to examine how linguistic patterns appear and whether they converge meaningfully across speakers and time.

Corpus Introduction

The Corpus, or data set, was sourced from the Fund for Independence in Journalism and is described as follows:

> “Researchers at the Fund for Independence in Journalism sought to **document every public statement made by eight top Bush administration officials from September 11, 2001, to September 11, 2003,** regarding (1) Iraq’s possession of weapons of mass destruction and (2) Iraq’s links to Al Qaeda. Although both had been frequently cited as rationales for the U.S. war in Iraq, by 2005 it was known that these assertions had not, in fact, been true” (Lewis et al.).

This project only utilizes text from primary sources that is specifically coded as relating to Iraq and or WMD. Each document is then coded as accurate or inaccurate based on the original data set’s definition of “false”, which is derived from the eventual finding that WMD were not present in Iraq and is explained as follows:

> “False statements by the eight Bush administration officials were…included in the total count of false statements—when they specifically linked Iraq to Al Qaeda or referenced Iraq’s contemporaneous possession, possible possession, or efforts to obtain weapons of mass destruction (chemical, biological, or nuclear weapons). In addition, any use of the verb “disarm” was categorized as a direct statement because of the literal meaning of the word. (Example: “Saddam Hussein has got a choice, and that is, he can disarm.”)” (Lewis et al.).

Time Period Definition

Each document in the data set is assigned to one of three time periods. T0: Early Threat Recognition; T1: Public Negotiation/Support Building (often referred to as the "lead-up" period just before the invasion of Iraq); and T2: The March 2003 Invasion of Iraq. These dates were selected for their historical significance based on well-reviewed and confirmed timelines, cited from the Council on Foreign Relations.

T0: Early Threat Recognition (September 11, 2001 – October 16, 2002)

- Post-9/11 documents link terrorism with WMD proliferation.
- The Bush administration increases mentioning of Iraq as a primary threat.
- Congress debates and passes the Authorization for Use of Military Force Against Iraq on October 16, 2002.

T1: Public Negotiation/Building Support (October 17, 2002 – March 19, 2003)

- Diplomatic appeals to the international community, such as former Secretary of State Colin Powell's UN Security Council presentation on February 5, 2003.
- Military planning and deployment intensifies.
- Congressional and intelligence hearings increase oversight.

T2: Operation Iraqi Freedom (March 19, 2003 – September 11, 2003)

- On March 19, 2003, President Bush addressed the nation on the situation with Iraq, posing an ultimatum to Saddam Hussein: "Our Nation enters this conflict reluctantly. Yet our purpose is sure. The people of the United States and our friends and allies will not live at the mercy of an outlaw regime that threatens the peace with weapons of mass murder" (Bush). Operation Iraqi Freedom begins.

Document Distribution Across Time Periods

Total documents for analysis: *n* = 536

- T0: *n* = 170
- T1: *n* = 189
- T2: *n* = 177

LIWC-22 Overview

This project uses LIWC-22 to score each individual transcript across a range of linguistic categories, each evaluated as a percentage of total words matching that category's subdictionary: "the LIWC-22 text processing module works by counting all of the words in a target text, then calculating the percentage of total words that are represented in each of the LIWC subdictionaries" (Pennebaker et al. 3).

The core variables selected for analysis are Analytic, Clout, Authentic, moral, and power. LIWC-22 defines these as follows (Pennebaker et al. 11):

- **Analytic** — metric of logical, formal thinking
- **Clout** — language of leadership and status
- **Authentic** — perceived honesty and genuineness
- **Tone** – degree of positive (negative) tone
- **moral** —most frequently used examples: wrong, honor*, deserv*, judge

- **power** — language associated with dominance and authority

It is important to note that all of the capitalized variables are weighted combinations of multiple subcategories, chosen for their coverage of the wider LIWC variable set (Pennebaker et al. 7). Moral and power were included for their contextual significance in political speech. For instance, the subcategory "certitude" contributes to Clout, while Authenticity incorporates personal pronoun use and words related to self-monitoring. A highlighted depiction of words matching variables that contribute to Authenticity for a specific text in the corpus is shown below:

Note that this document has high Authenticity (score = 88.72/100). Color coding is as follows (this is not a comprehensive list of all contributing Authenticity subgroups, but an example of how LIWC-22 works):

- Green — words relating to 'insight'
- Purple — 'I'/personal pronouns
- Orange — words relating to 'differing'

I think the president made it very clear today that this is about **more than** just one organization, it's about **more than** just one event, as horrible as that event was. We are engaged in a war **against** terrorism, **against** the terrorist networks, **against** state support for terrorism. And **I think** everyone has got to look at this problem with completely new eyes in a completely new light after what happened last Tuesday. **But if** you go back to what the president said just earlier today, when he was greeting President Megawati of Indonesia, he made it very clear that this is about **more than** just one country **or** just one individual **or** just one organization. You **know** that **I** don't **answer questions** like that.

Categorization Methods

Documents are sorted into the following categories: speeches, press briefings, interviews, and official documents. As recommended by LIWC22, prepared speech (speeches, official documents) is analyzed separately from spontaneous speech (press briefings and interviews). Note that for Authenticity, the most significant variable across time periods, prepared speech cannot be sufficiently analyzed by LIWC, so spontaneous speech is the focus. For any document with multiple speakers, only speech from the primary individual (i.e., Ari Fleischer during a press briefing) is retained.

Key speakers are selected based on their continuity through or frequency in key sections of the corpus for best significance. The key speakers are as follows:

- President George W. Bush
- Press Secretary Ari Fleischer
- Secretary of State Colin Powell
- Secretary of Defense Donald Rumsfeld
- Deputy Secretary of Defense Paul Wolfowitz
- Vice President Richard (Dick) Cheney

Initial Test for Variable Normality:

Shapiro-Wilk's test rejects normality for all six core LIWC variables in both document types (all $p < .05$, most $p < .0001$) (Ghasemi and Zahediasl). So, non-parametric tests for significance are used throughout the project. 'Moral' is the most severely right-skewed variable, note it is not a key variable, (skew = 2.49 spontaneous, 2.02 prepared), followed by power (skew

= 1.37 prepared) and tone (skew = 1.33 spontaneous). However, they are not the most significant variables in the project.

Collinearity Amongst Core Variables:

Among the core variables, the most relevant correlation (Spearman's for non-parametric) is between Clout and Authenticity (prepared: $\rho = -0.52$; spontaneous: $\rho = -0.32$), which is addressed by the depiction of Clout as a corroborating variable (Schober et al.). Since all inferential tests are applied to single variables independently, collinearity does not affect their validity. See the collinearity matrices in Figure 1 below:

Figure 1

PREPARED :

	Analytic	Clout	Authentic	Tone	moral	power
Analytic	1.00	-0.01	-0.18	-0.18	0.20	0.54
Clout	-0.01	1.00	-0.52	-0.15	0.13	0.13
Authentic	-0.18	-0.52	1.00	0.19	-0.19	-0.37
Tone	-0.18	-0.15	0.19	1.00	-0.10	-0.40
moral	0.20	0.13	-0.19	-0.10	1.00	0.23
power	0.54	0.13	-0.37	-0.40	0.23	1.00

SPONTANEOUS :

	Analytic	Clout	Authentic	Tone	moral	power
Analytic	1.00	0.06	-0.31	-0.09	0.08	0.38
Clout	0.06	1.00	-0.32	-0.12	0.07	0.09
Authentic	-0.31	-0.32	1.00	0.25	-0.14	-0.46
Tone	-0.09	-0.12	0.25	1.00	-0.06	-0.43
moral	0.08	0.07	-0.14	-0.06	1.00	0.13
power	0.38	0.09	-0.46	-0.43	0.13	1.00

Significant Variables

Authenticity and Clout were the only two core variables to show significant variation across time periods in spontaneous documents (Kruskal-Wallis test for Authenticity: $H(2) = 11.53$, $p = .003$, $\varepsilon^2 = .032$; Clout: $H(2) = 6.45$, $p = .040$, $\varepsilon^2 = .015$), both with small effect sizes. All other variables were non-significant. Authenticity was the only significant variable in prepared speech, though that finding is far less relevant due to the naturally inauthentic nature of pre-written speech. As distribution shapes differ across time periods (Figure 3), KW results are interpreted as tests of stochastic dominance rather than strict median difference (Dinno).

Both variables survived Dunn-Bonferroni post-hoc correction between T1 and T2 (Authenticity: statistic = 3.48, p.adj = .002; Clout: statistic = 2.51, p.adj = .036), though the effect was stronger for Authenticity. The Bonferroni adjustment multiplies each p-value by $m = k(k-1)/2 = 3$, where $k = 3$ time periods (Dinno). No other time period comparisons survived correction for either variable.

Clout is considered a reinforcing variable rather than a key finding. Though inversely correlated with Authenticity at the document level ($\rho = -0.32$), median Clout followed the same trajectory across time periods (Figure 2), suggesting the two dimensions follow a shared linguistic trend during the build-up period, with their negative correlation stemming from within-period variation rather than differing patterns over time.

Primary Finding: T1 Authenticity Drop

T1 is therefore identified as the period of lowest Authenticity. Authenticity was significantly higher in T2 than T1. The decrease from T0 to T1 is consistent with this pattern but does not independently reach significance. This is interpreted as stochastic dominance, which is the statistically significant probability that a randomly drawn document from T2 will have higher Authenticity than one from T1 (Dinno).

Figure 2

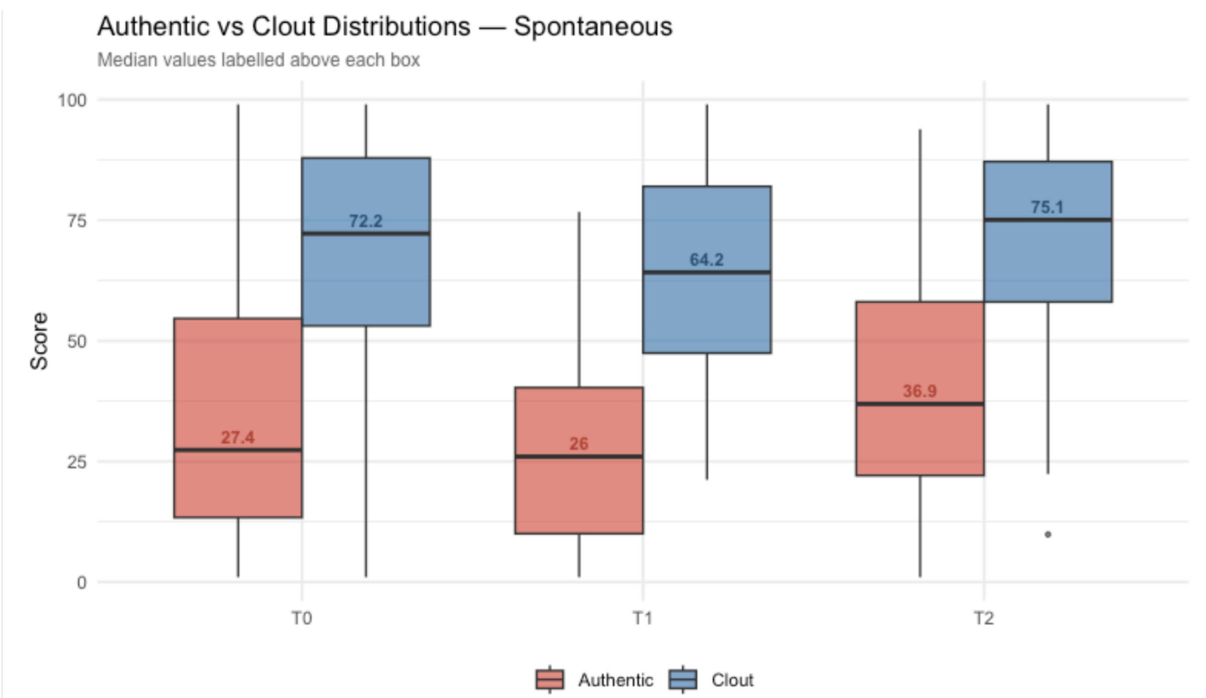
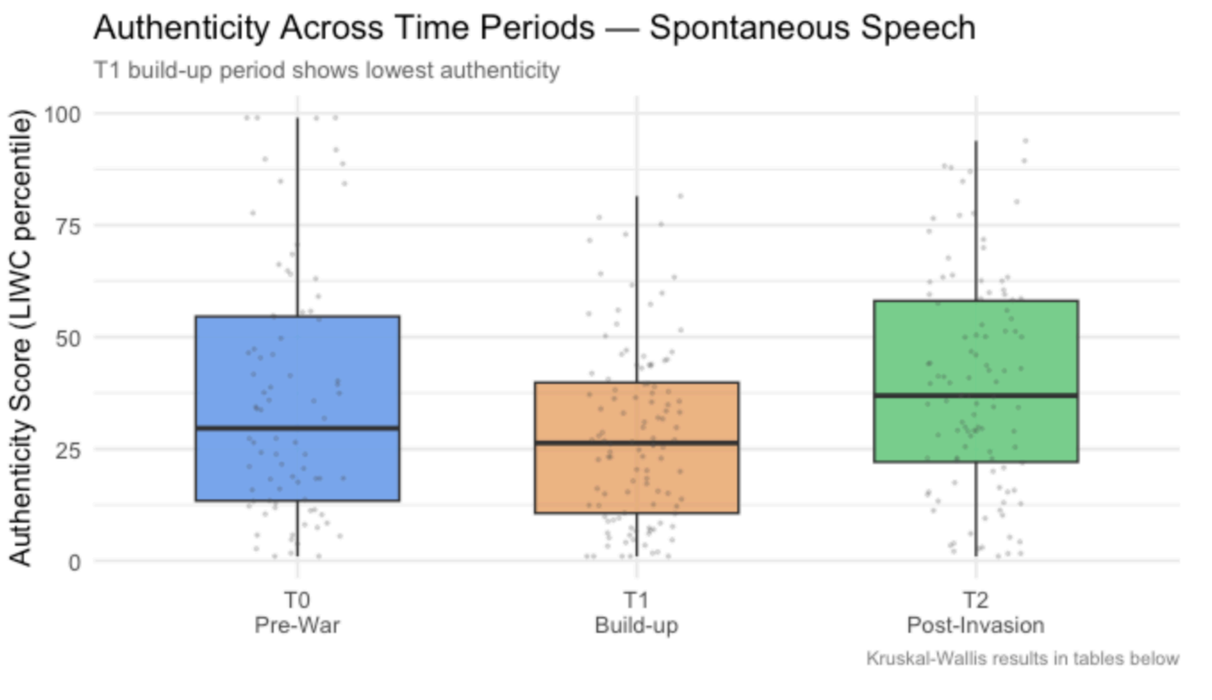


Figure 3



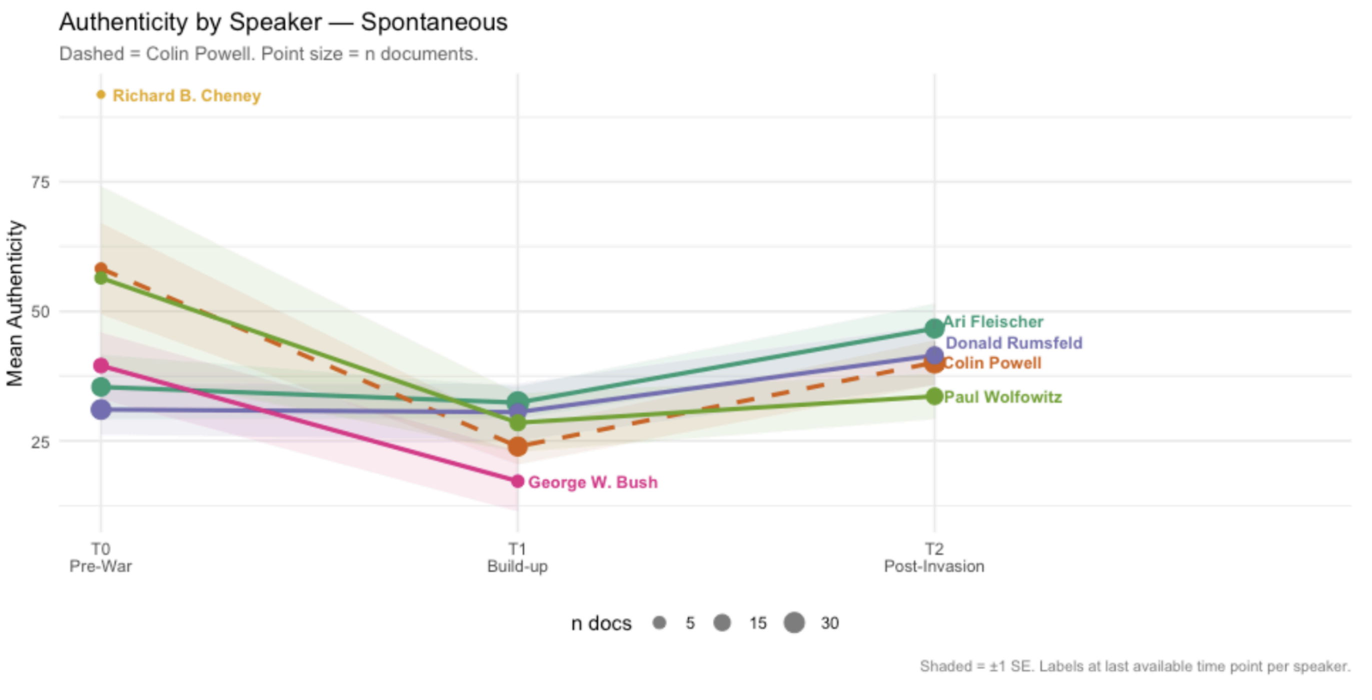
Word Count Sensitivity Check

LIWC22 Suggests that word count be greater than or equal to 100 (Pennebaker et al. 35). Since some of our documents are under 100, we test to see whether word count has a significant effect on Authenticity (the primary variable for analysis). We find that it doesn't, as the Spearman correlation between WC and Authenticity is non-significant in both document types: prepared $r = .134$, $p = .060$; spontaneous $r = .077$, $p = .192$.

Also, the T1 to T2 Dunn-Bonferroni pairwise correction is still significant when run for word count ≥ 100 documents only: $H(2) = 15.8$, $n = 244$, $p < .001$.

Individual Speaker Mean Trajectories and Sign Test

Figure 4



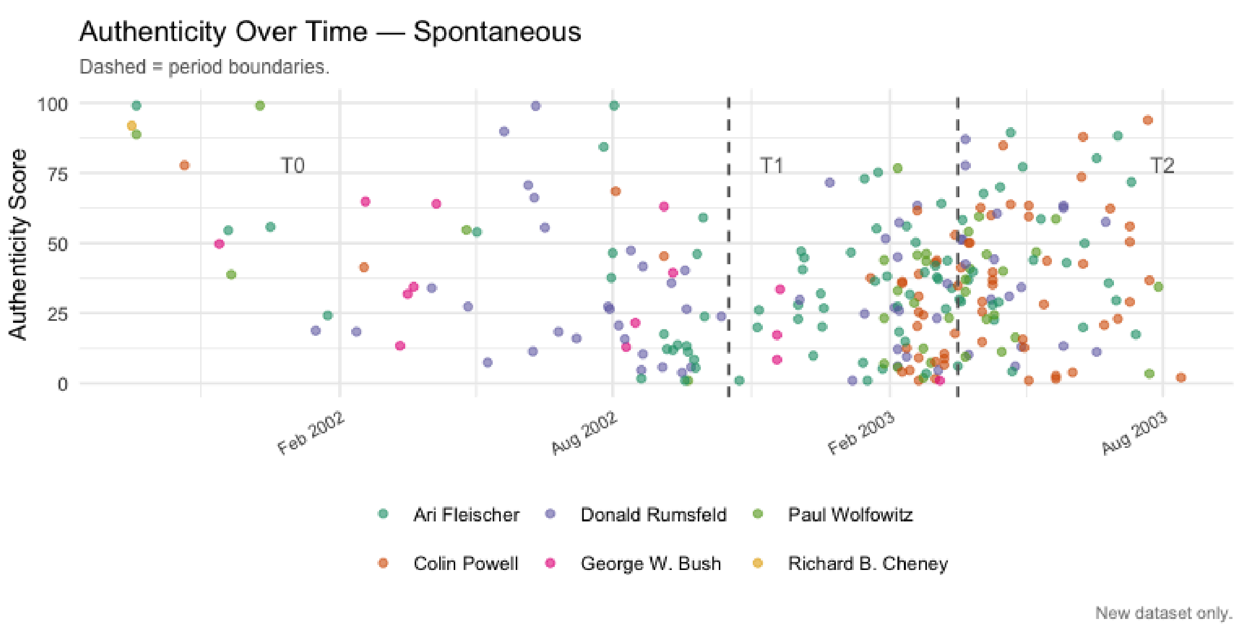
In spontaneous documents, all 5 key speakers show a T0 to T1 drop (Figure 4). It is important to note that speakers operated within a related context, so transcripts are not completely independent observations. Also, speakers do not have an even distribution of documents in each time period, some (note President W. Bush or V.P. Cheney) do not have spontaneous documents in all of the time periods. Speaker-level comparisons are meant to show that no individual is responsible for the trend in Authenticity over time and are illustrative only.

Document-Level Authenticity across Timeline:

The scatter plot of individual documents across the timeline contextualizes the T1 authenticity drop (see Figure 5). In T0, documents are more sparsely distributed across time with scores across the full range. After T0, there is a higher density of observations clustering toward lower values, especially just before the invasion. Still, the trend is visually consistent. This suggests the drop reflects a shift in the style and frequency with which figures were speaking, not just a reaction to outliers.

The later periods are more dominated by Fleischer, Rumsfeld, and Powell. President Bush and V.P. Cheney have few spontaneous documents, and their contributions to T0 tend toward higher scores. Their reduced presence in later periods could partially account for the drop. Still, speaker-level Authenticity means follow the same directional pattern across time periods, suggesting the pattern holds regardless of who is speaking.

Figure 5



Speaker Norm Comparisons

All spontaneous speakers are significantly below the LIWC22 Authenticity mean for conversation (M = 69.56), with z-scores ranging from −1.86 (Bush) to −1.62 (Fleischer). Cheney is excluded due to a lack of spontaneous documents.

The plots below show overall deviation of core variables from LIWC22 norms with prepared documents compared against speech norms, taken from US Congressional speeches, and spontaneous against conversation norms (Pennebaker et al. 9). The standard-deviations do not assume normality, but are taken directly as reported in the LIWC22 norms data set (*LIWC-22 Descriptive Statistics*). (Figures 6 and 7).

Deviation from norms is wider for spontaneous speech than prepared speech, indicating that press briefings and interviews are more different from typical conversation than prepared speeches are from typical congressional speech. This is not surprising given differences in context. Authenticity is the only variable consistently below its norm in spontaneous speech, indicating more self-monitoring, formalized rhetoric than in typical conversation. It is worth noting, however, that even the greatest deviation (spontaneous Authenticity at T1) remains fewer than 3 standard deviations from the conversation norm. Though Authenticity shows significant patterns within the data set, the documents in the corpus are not significantly inauthentic on average.

Regarding moral language, President George Bush's prepared speech mean (M = 0.297) is the highest among key speakers, though all speakers fall below the LIWC speech norm of 0.40. His period-level means (T0: M = 0.302, T1: M = 0.330, T2: M = 0.222) show a slight elevation in T1 that drops sharply post-invasion. Medians are zero across all periods however, and the severely right-skewed distribution indicates that moral language was most present in a small group of speeches.

Figure 6

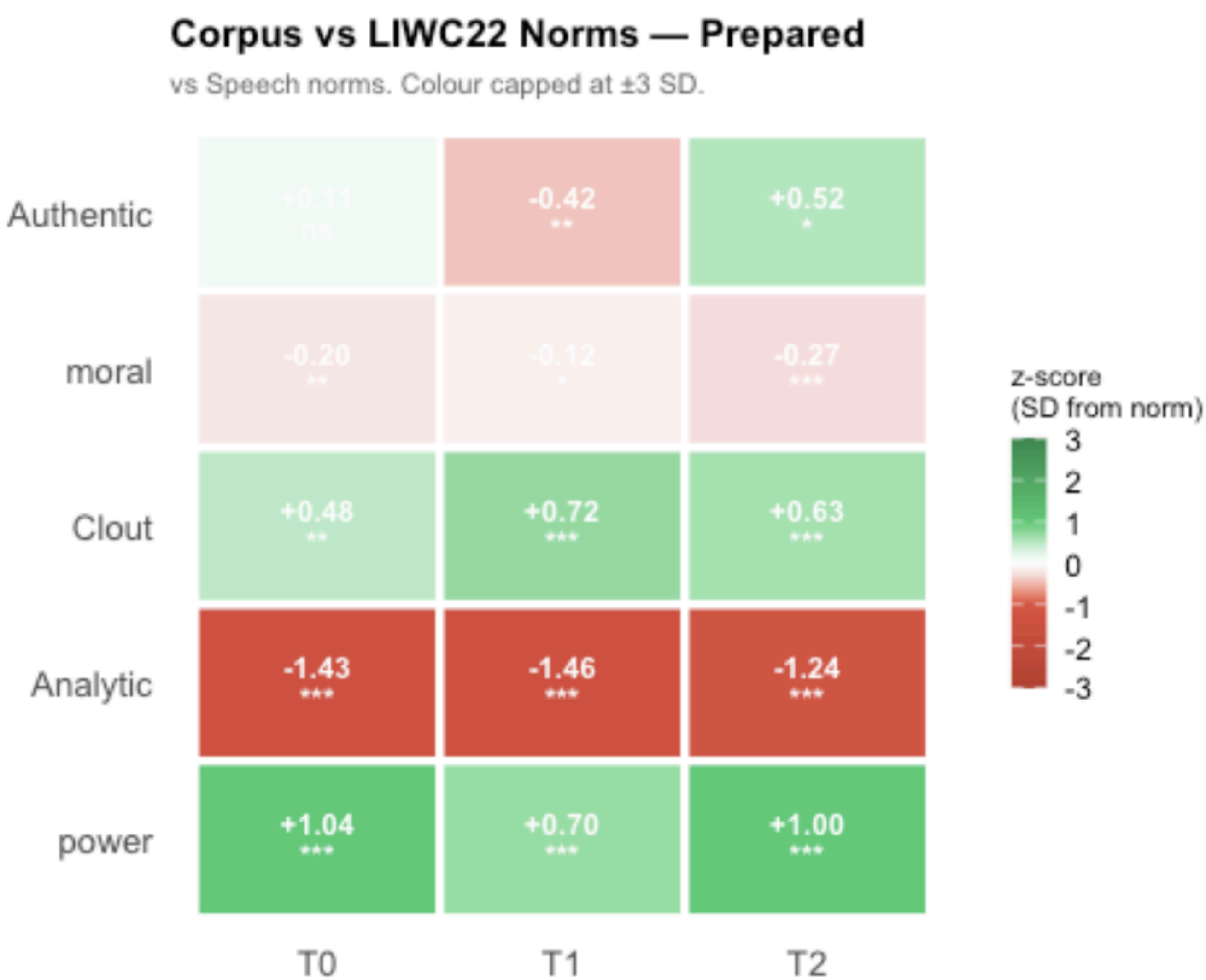
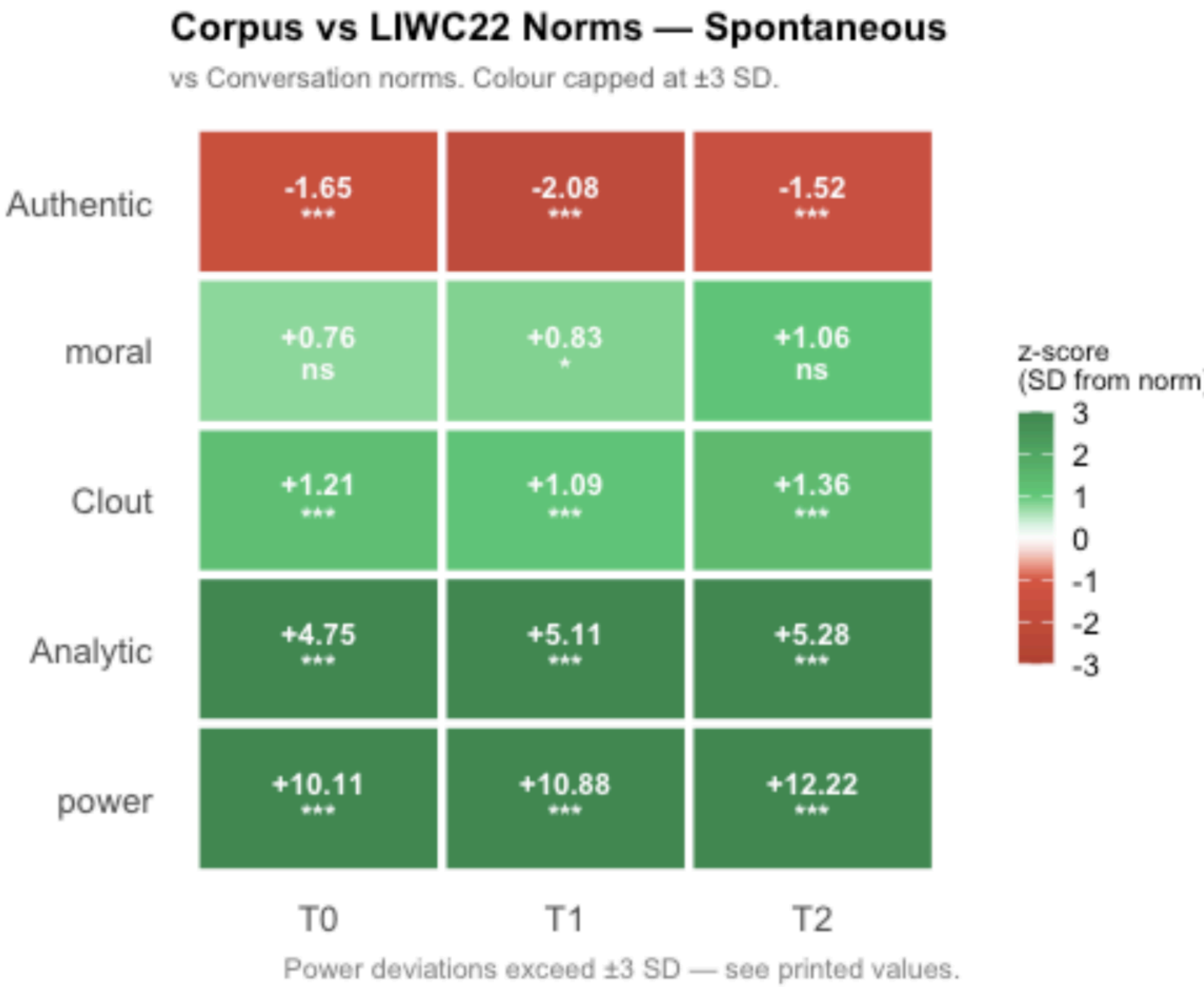


Figure 7


Corpus vs LIWC22 Norms — Spontaneous
vs Conversation norms. Colour capped at ±3 SD.
Authentic
moral
Clout
Analytic
power
-1.65 ***
-2.08 ***
-1.52 ***
+0.76 ns
+0.83 *
+1.06 ns
+1.21 ***
+1.09 ***
+1.36 ***
+4.75 ***
+5.11 ***
+5.28 ***
+10.11 ***
+10.88 ***
+12.22 ***
T0
T1
T2
z-score
(SD from norm)
3
2
1
0
-1
-2
-3
Power deviations exceed ±3 SD — see printed values.

Random Forest: LIWC as Predictor of Claim Accuracy

A Random Forest classifier uses multiple decision trees in a “voting system” to test the accuracy of non-parametric predictors. It is also a useful tool for correlated predictors, which are present in this variable set (Strobl et. al). This model tests whether the core LIWC dimensions carry independent predictive signals for whether a document is coded as accurate or inaccurate, after accounting for the predictive power of time period and speaker.

To test variables as predictors of accuracy (again, disputed (or in-accurate) documents assume the presence of WMD), a two-model Random Forest comparison was run separately for prepared (n = 175; 16 accurate, 159 disputed) and spontaneous (n = 255; 65 accurate, 190 disputed) documents, with class weighting applied due to imbalance in sample sizes between binary classifiers. Variable importance was also estimated using permutation importance with bootstrap confidence intervals (n = 300 resamples), which tests whether the CI of a given predictor contains 0 (no contribution) or a negative value (model degradation) after it is generalized for simulated data (Strobl et. al). Then, a Receiver Operating Characteristic Curve is produced from either model. AUC (Area Under Curve, which reports the probability of correct ranking) is reported as the primary metric given class imbalance (Hanley and McNeil).

The base model for comparison includes speaker and time period only, since both show notable effects on core variables, as well as a clear relationship to accuracy. The model showed adequate predictive accuracy in both document types (prepared: AUC = .717; spontaneous: AUC = .738 with a threshold of 0.7), confirming that accurate claims are not randomly distributed across categorical variables.

Adding LIWC (continuous) variables degraded the performance of the base model:

- Adding 6 Core Variables: prepared AUC = .648 (−.069); spontaneous AUC = .701 (−.037)
- All recorded LIWC variables (17): prepared AUC = .590 (−.127); spontaneous AUC = .616 (−.122)

The lower performance of both expanded models indicates that LIWC dimensions together do not strengthen prediction and introduce noise. However, multicollinearity may play a role, so each core variable is also tested in its own individual extension of the base model.

Individual core variables showed small improvements to the base model AUC (Figures 9 and 11):

- Prepared: Power (+.063), Analytic (+.044), Clout (+.031)
- Spontaneous: Power (+.023), Clout (+.022), Authentic (+.001)

Note, the prepared data has a larger sample size imbalance between accurate and inaccurate, and should be interpreted with higher caution.

The following Figures show the AUC by model as compared to the base model as well as the boot-strapped Importance of each core variable as an independent addition. We see that power and Analytic show highest bootstrap importance for prepared documents (Figure 10) and power and Authenticity (though Authenticity's AUC gain is very small) are highest for spontaneous documents. (Figure 8)

Figure 8

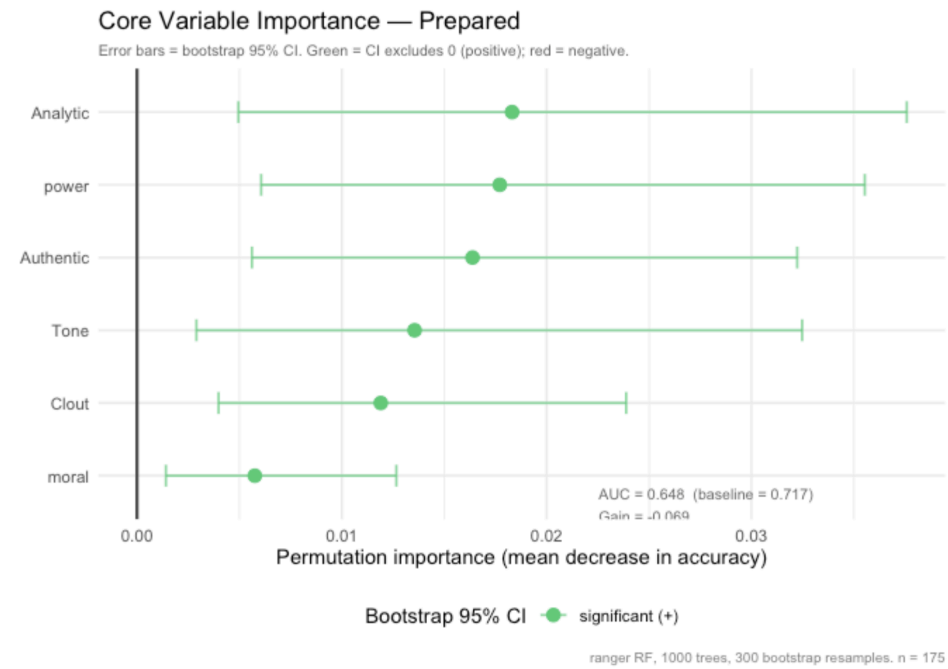


Figure 9

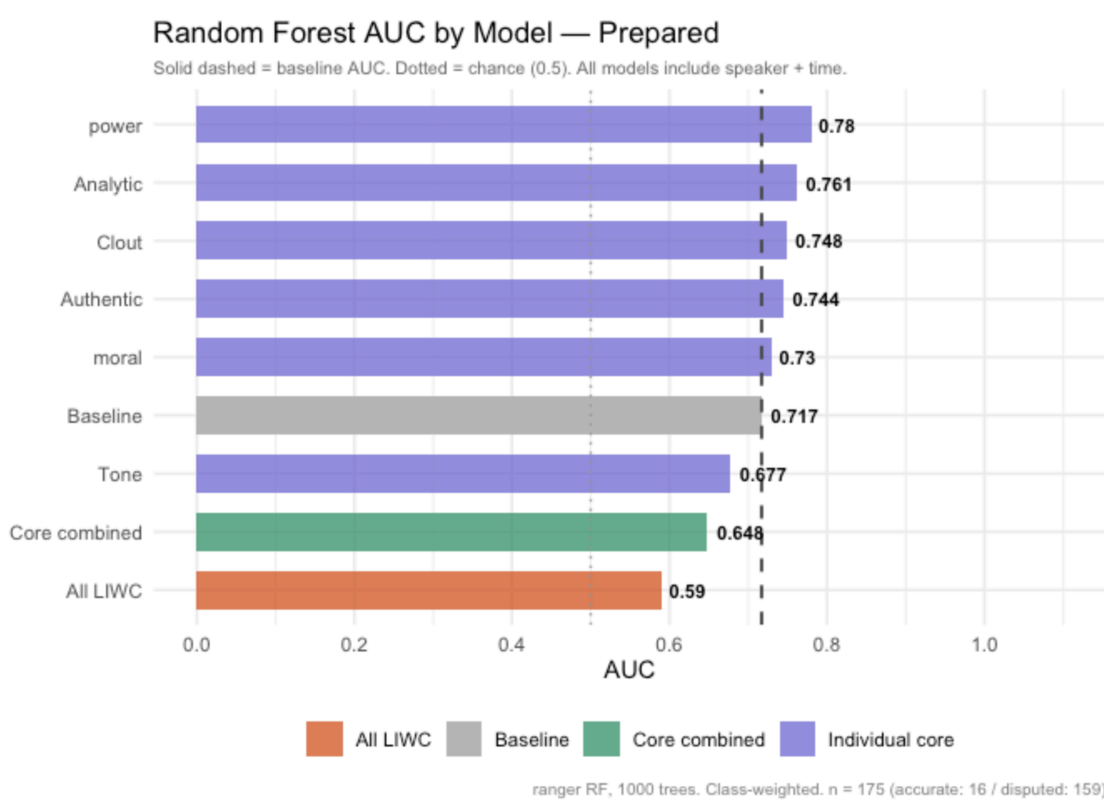


Figure 10

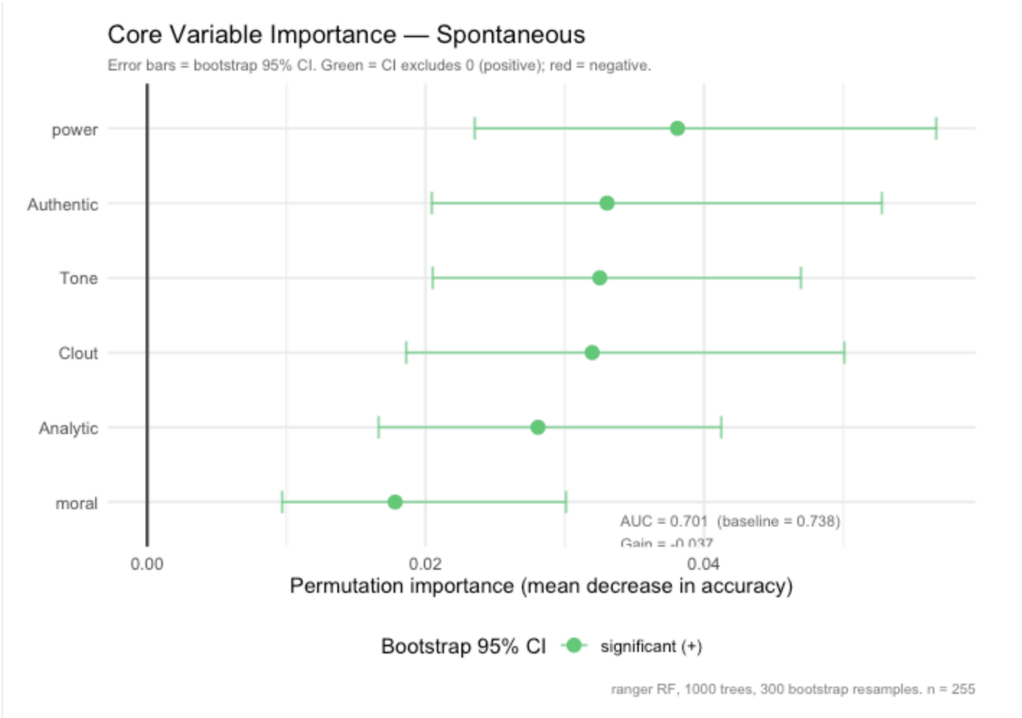
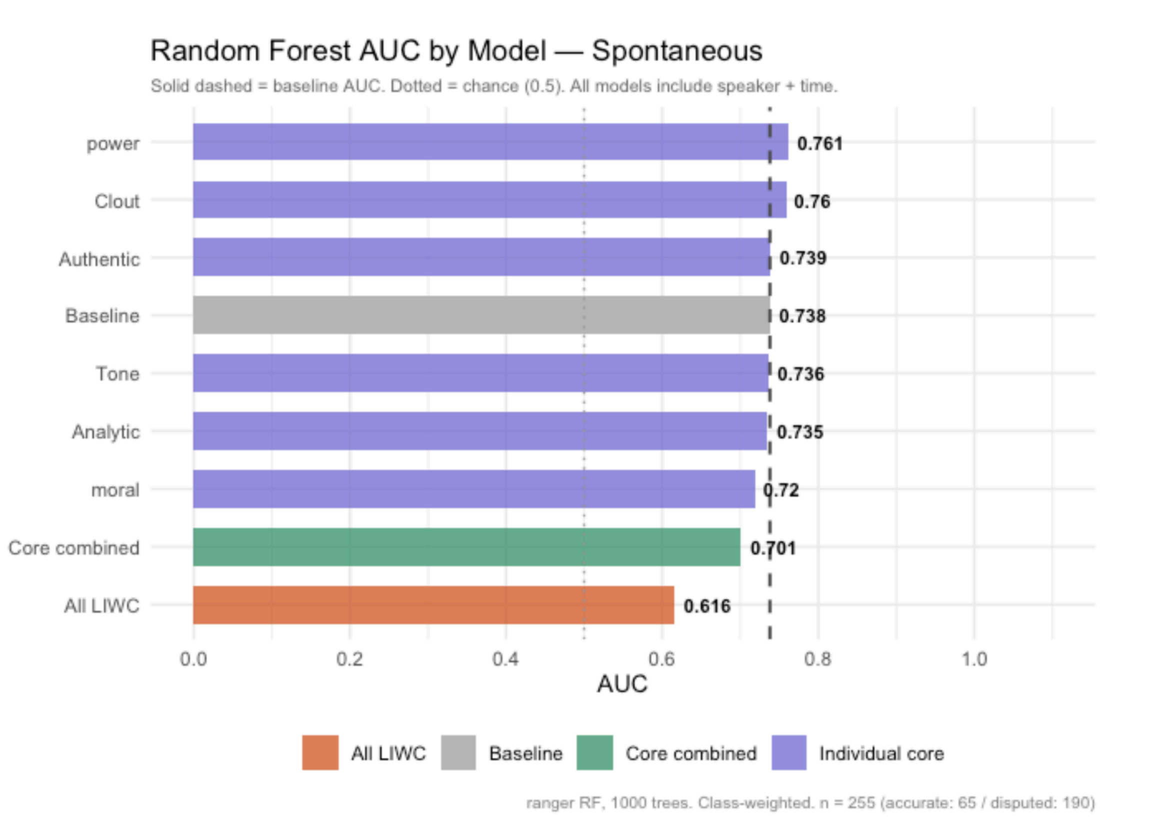


Figure 11



Conclusion

Findings support that spontaneous, public-facing communication among key administration officials shows a significant downward shift in Authenticity and Clout across the lead-up to the 2003 invasion of Iraq. Patterns in Authenticity are stronger, surviving a more conservative statistical correction and showing directional consistency across all speakers with an adequate sample size. Mapping individual document Authenticity scores across the timeline confirms that these trends are not driven by outlying instances but by consistent patterns in the data, with some clustering around historically significant boundary dates— The AUMF and the invasion.

Core LIWC variables do not independently predict claim accuracy once speaker and time period are accounted for, though Power and Clout show modest individual improvements across both document types, with Analytic notable for prepared speech. All improvements are small, supporting the interpretation that who was speaking and when are more indicative of claim accuracy than linguistic style alone. These findings contradict the idea that rhetorical patterns, as measured by LIWC, relate strongly to inaccuracy in this case.

In short, there are measurable convergences in linguistic registers across time. These changes are significant in Authenticity and corroborated by Clout, but they do not predict accuracy, suggesting the observed patterns reflect institutional shifts in expressive style rather than markers of inaccurate or exaggerated communication.

Limitations

Due to limitations in collecting, cleaning, and coding an adequate number of transcripts, findings cannot be generalised beyond this specific case. However, comparison to LIWC norms offers some reference to typical speech. For further study, analysis of other conflicts is recommended to test whether timeline convergence patterns are similar.

Speakers are not fully independent actors. Due to their shared institutional and historical context, their rhetorical choices can be reasonably assumed to affect one another, complicating independence assumptions.

There is an uneven distribution of documents across time periods, though sample size remains viable for significance testing overall. Within individual speakers, n is not large enough to test time period significance, with some speakers retaining no documents of a certain type in certain time periods. Speaker-level observations are therefore reported as informative rather than statistically significant findings.

Class imbalance in the Random Forest models limits the reliability of the results, though this is partially mitigated by class weighting and bootstrapped confidence intervals.

LIWC-22, though often used for political speech, is dictionary-based, and linguistic patterns specific to the time period and context may not be covered by the dictionaries. For further study, a more customized model is recommended.

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